Stanley Okwii

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Summary

Senior Software Engineer with 7+ years building production systems in **Python** and **TypeScript** on **AWS**, across payments, analytics products, and data pipelines. Currently founding engineer on an AI research assistant, owning the hybrid retrieval pipeline, agent orchestration, and multi-tenant cloud infrastructure. Experienced across the full lifecycle from schema design and IaC to **React** frontends, leveraging modern agentic workflows for rapid delivery while maintaining strict ownership of system architecture, security, and reliability.

Skills

Programming: Python, **TypeScript**, JavaScript, SQL

AI & ML: RAG pipelines, hybrid retrieval, **pgvector**, embeddings, LLM integration (**Claude**, Gemini), prompt engineering

Backend: Django, **Django REST Framework**, **FastAPI**, **Node.js**, **Celery**, Redis Streams

Frontend & Mobile: React, **Next.js**, **Flutter**

Databases: PostgreSQL, **Redis**, **MySQL**, MongoDB

Cloud & DevOps: AWS (ECS Fargate, Bedrock, CloudFormation), **Docker**, **Terraform**, **GitHub Actions**, CI/CD, Sentry, New Relic

Testing: Pytest, **Playwright**, Jest

Developer Tooling: Claude Code (custom skills & agent workflows), Dev Containers

Experience

Founding Engineer, Access to Energy – London, UK Jan 2026 – present

An AI research assistant that answers questions on energy access in Africa, citing evidence from a curated library of sector reports. Team of 4. I own the backend, the AI pipeline and the infrastructure.

- Built the core **hybrid search** engine combining keyword matching (**BM25**) and semantic vector search (**pgvector**, HNSW) in **PostgreSQL**, delivering accurate passage retrieval without external database overhead.
- Built the end-to-end **RAG** assistant in **Python** with a custom, lightweight pipeline, routing questions to **AWS Bedrock** (**Claude**) with streaming inline citations and automated Gemini fallback.
- Built the asynchronous document ingestion pipeline using **Redis Streams** workers (consumer groups, dead-letter retries) to parse, chunk, and embed large PDFs without degrading API response times.
- Designed **multi-tenant** data isolation and role-based access controls in **PostgreSQL**, letting multiple client organisations share one deployment while strictly partitioning documents.
- Automated **AWS** infrastructure provisioning using modular nested **CloudFormation** stacks to deploy **ECS Fargate** microservices, ElastiCache **Redis**, and ALB, reducing environment setup from days to minutes.

Full Stack Software Engineer (Contract, Part Time), Hoppers – Remote Oct 2025 – present

Event ticketing platform (hoppers.ug). Built the multi-provider payment engine and high-concurrency reservation system.

- Built the multi-provider payment engine (DPO, IoTec), implementing idempotent webhook handling via row-level locking to eliminate duplicate charges.
- Prevented ticket overselling during demand spikes by building a high-concurrency reservation engine with guarded atomic updates in **Django** and database check constraints.
- Enforced purchase limits and venue table reservations by serializing booking state transitions with row-level locks, eliminating checkout race conditions.
- Delivered core platform features end-to-end across the **Django** backend, **React** admin panel, and mobile app, covering QR code ticket verification and role-based permissions.
- Built an agentic development environment across isolated git worktrees and Dev Containers with dynamic port allocation, automating migrations, seed data, and pull requests verified by **Playwright** UI diffs and replayed API tests across web and **Flutter** apps.

Senior Full Stack Software Engineer, Harver – New York, USA (remote) Mar 2023 – May 2025

Promoted to Senior in March 2023. Stepped away in May 2025 to complete MSc in Data Analytics full time.

- Owned the architecture and delivery of three enterprise client products covering candidate assessment matching, recruiter portals, and analytics dashboards, built with **Django REST Framework**, **PostgreSQL**, and **React**.
- Led technical design for core platform features, defining data models and API contracts while building asynchronous processing queues with **Celery** and **RabbitMQ**.
- Resolved high-priority production incidents for enterprise clients, using **New Relic** and **Sentry** to trace complex bugs across distributed services and ship permanent fixes.
- Ran cross-functional project planning, breaking complex product roadmaps into phased milestones and managing risk mitigation to hit scheduled release dates.
- Conducted 15+ technical interviews for system design and code quality, and authored onboarding documentation for core services to accelerate new hire ramp-up.

Full Stack Software Engineer, Harver – New York, USA (remote) Sept 2019 – Mar 2023

- Introduced structured logging and APM monitoring across distributed **Django** services with **Sentry** and **New Relic**, replacing speculative debugging with direct error-to-cause tracing.
- Built talent mobility dashboards in **React**, enabling enterprise clients to track employee career progression and skills across their organizations.
- Profiled high-latency API endpoints, eliminating N+1 database queries and implementing **Memcached** caching on hot data paths to cut response latency.
- Automated deployment pipelines with **GitHub Actions** and **Docker**, replacing manual deployment procedures with one-click CI/CD releases.

Mentorship

CareerFoundry (Online Coding Bootcamp – Berlin, Germany Mar 2023 – Jan 2026

Full Stack Web Development Mentor (Freelance)

- Mentored 23+ students through the Full Stack Web Development program and held a 5/5 student rating for the quality and speed of my feedback.

Education

MSc **Birkbeck, University of London**, Data Analytics – London, UK Oct 2024 – Sept 2025

- Merit. Dissertation on adversarial robustness in image classification: implemented PGD attacks from scratch and adversarially retrained ResNet50 on an ImageNet subset.

BSc **Makerere University**, Software Engineering – Kampala, Uganda Aug 2014 – Sept 2018

- 2:1 (Upper Second Class Honours)